

PRACTICE GUIDANCE

AASLD Practice Guidance on the use of TIPS, variceal embolization, and retrograde transvenous obliteration in the management of variceal hemorrhage

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PURPOSE AND SCOPE OF THE GUIDANCE

This is a comprehensive guidance on the use of interventional radiology endovascular techniques in the management of variceal hemorrhage from the American Association for the Study of Liver Diseases (AASLD). This guidance document is complementary to the AASLD “Risk Stratification and Management of Portal Hypertension and Varices in Cirrhosis” guidance^[1] and addresses the recent advancements in these invasive procedures. Although the use of TIPS

dates back to the 1980s, several new technical refinements in TIPS stents have occurred in the last few years. The other major addition to the management of gastric variceal hemorrhage in North America has been the introduction of retrograde transvenous obliteration (RTO) in its different forms. The present document aims to equip care providers with an in-depth understanding of the use of TIPS and/or variceal embolization/obliteration in the management of variceal hemorrhage. The goal is to facilitate multidisciplinary discussions between hepatologists, gastroenterologists, interventional radiologists, and surgeons in the

Abbreviations: AASLD, American Association for the Study of Liver Diseases; ATO, anterograde transvenous obliteration; BRTO, balloon-occluded RTO; CARTO, coil-assisted RTO; CT, computed tomography; CTP, Child-Turcotte-Pugh; GRS, gastroduodenal shunt; ICU, intensive care unit; mBRTO, modified BRTO; NSBB, nonselective beta-blocker; PARTO, plug-assisted RTO; PSPG, portosystemic pressure gradient; PTFE, polytetrafluoroethylene; RCT, randomized controlled trial; RTO, retrograde transvenous obliteration; US, ultrasound.

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selection of endovascular treatments for patients with variceal hemorrhage.

This document reviews first the anatomy of portosystemic collaterals, which is essential to understand the rationale for the use of these endovascular procedures. A second section describes the technical details of TIPS, RTO, and antegrade transvenous obliteration (ATO). The final section deals with the clinical use of endovascular treatments in the management of hemorrhage from esophageal, gastroduodenal, and ectopic varices.

Therefore, this AASLD Guidance provides a data-supported approach to the use of endovascular therapies for the management of variceal hemorrhage. It differs from the AASLD Guidelines, which are supported by systematic reviews of the literature, formal rating of the quality of the evidence, and strength of the recommendations. In contrast, this Guidance was developed by consensus of an expert panel and provides guidance statements based on comprehensive review and analysis of the literature on the topics, with oversight provided by the AASLD Practice Guidelines Committee. The AASLD Practice Guidelines Committee chose to perform a Guidance on this topic because, in most instances, a sufficient number of randomized controlled trials (RCTs) were

not available to support meaningful systematic reviews and meta-analyses on this topic.

ANATOMY OF PORTOSYSTEMIC COLLATERALS IN PORTAL HYPERTENSION

Portosystemic collaterals develop between the portal and the systemic circulation as a consequence of portal hypertension. They represent an unsuccessful attempt by the body to decompress the portal circulation through the systemic circulation. Collaterals in the gastrointestinal lumen are referred to as varices. Varices develop commonly at the distal esophagus and proximal stomach i.e., the gastroesophageal region. Varices outside the gastroesophageal area such as the rectum, duodenum, and surgically created sites (e.g., stomal) are uncommon ($\leq 5\%$ of all variceal bleeding) and are referred to as ectopic varices.^[2–4] Table 1 shows the location, supply, and drainage of common varices.

Esophageal varices are located in the mucosa and submucosa of the mid- and distal esophagus. They are supplied by the left gastric vein and drain into the azygos and hemiazygos veins through periesophageal

TABLE 1 Location, supply, and drainage of gastrointestinal varices (see also Fig. 1)

Location		Supply (afferent)	Drainage (efferent)
Gastroesophageal varices			
Esophageal varices	Mid- and distal esophagus	Left gastric vein	Through azygos and hemiazygos vein into SVC
Gastroesophageal varices (GOV): contiguous with esophageal varices			
GOV1	Lesser curvature of stomach	Left gastric vein	Through azygos and hemiazygos veins into SVC
GOV2	Greater curvature of stomach/fundus	Left gastric vein and posterior gastric vein	Through azygos vein and hemiazygos veins into SVC and inferior phrenic vein into left renal vein (GRS) or IVC (gastrocaval shunt)
Isolated gastric varices (IGV)			
IGV1	Fundus	Short gastric veins, posterior gastric vein	Through inferior phrenic vein into left renal vein (GRS) or into IVC (gastrocaval shunt)
IGV2	Any location in stomach other than the fundus	Short gastric veins, posterior gastric vein	Left and right gastric vein
Ectopic varices			
Rectal varices	Rectum	Superior rectal vein	Through middle and inferior rectal vein into internal iliac vein
Duodenal varices	Duodenum	Superior or inferior pancreaticoduodenal veins ^[5,6]	Through gonadal or renal capsular vein into IVC
Stomal varices	Surgically created ostomies	Mesenteric branches from the superior mesenteric vein ^[7]	Through systemic venous collaterals in abdominal wall into iliofemoral vein

Abbreviations: GRS, Gastrorenal shunt; SVC, superior vena cava.

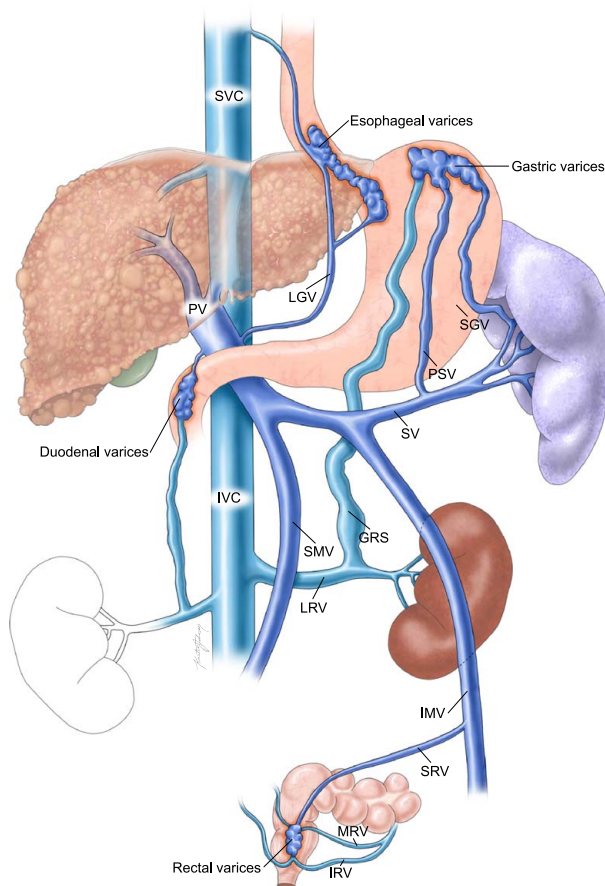


FIGURE 1 Supply and drainage of varices. Esophageal varices and GOV1 are supplied by the left gastric vein (LGV) and drain through the azygous vein (AV) into the superior vena cava (SVC). Gastrofundal varices are supplied by the LGV (not shown in the figure), posterior gastric vein (PGV), and/or short gastric vein (SGV). Gastrofundal varices can drain into the left renal vein (LRV) through a gastrosplenic shunt (GRS). Rectal varices are supplied by the superior rectal vein (SRV) and drain into the middle rectal vein (MRVs) and inferior rectal veins (IRV). Duodenal varices are supplied by the superior or inferior pancreaticoduodenal veins and drain into the IVC through the gonadal or the renal capsular vein.

veins (gastroesophageal venous system) (Fig. 1). Esophageal varices are most likely to bleed at the gastroesophageal junction because they are more superficial in this location and are unable to decompress through the azygos vein because of lack of communication with the periesophageal veins. Based on subjective endoscopic appearance, esophageal varices can be classified into (1) small: straight varices that collapse on insufflation, (2) medium: enlarged, tortuous varices that occupy less than one third of the lumen, or (3) large: coil-shaped varices that occupy more than one third of the lumen.^[8] In practice, medium and large varices are grouped together as high-risk varices because they require prophylactic therapy to prevent variceal bleeding.^[8] Owing to their superficial location, esophageal varices are amenable to endoscopic

variceal ligation (EVL), i.e., placing of rubber bands for variceal obliteration, or sclerotherapy.

Gastric varices can be grouped based on their endoscopic location as (1) gastroesophageal varices type 1 (GOV1): lesser curvature, (2) gastroesophageal varices type 2 (GOV2): greater curvature, (3) isolated gastric varices type 1: gastric fundus (IGV1), and (4) isolated gastric varices type 2: any location in the stomach other than the gastric fundus (IGV2).^[9] Gastric varices are supplied by the left gastric vein (feeds GOV1, GOV2), or short gastric veins, and posterior gastric vein (feed IGV1, IGV2, GOV2) (Fig. 1). Like esophageal varices, GOV1 and sometimes GOV2 drain into the gastroesophageal venous system (Fig. 1). GOV2 and IGV1, also known as “cardio-fundal varices,” drain through the gastrophrenic venous system, which is through the inferior phrenic vein into the left renal vein (gastrosplenic shunt [GRS]) or the IVC (gastrocaval shunt) (Fig. 1). These can be seen frequently in patients with portal and/or splenic vein thrombosis. The accompanying large gastrosplenic/caval shunts provide an opportunity for transvenous obliteration. Gastric varices tend to be submucosal and may not be amenable to EVL. From an interventional radiology perspective, several classifications of gastric varices have been proposed based on characteristics important for planning interventional radiology procedures such as the type of varices and presence of shunts,^[4,10] the anatomy of the feeding and drainage vessels,^[11] visualization during venography,^[12] and the flow dynamics of the circuit^[13] (Table 2). Regardless of the classification, intraoperative flow dynamics are most important in determining the technique of embolization.

IGVs (IGV1 > IGV2) can form in sinistral or segmental portal hypertension from a focal splenic vein thrombosis or stenosis related to an inflammatory process such as pancreatitis. Unlike in cirrhosis, in which IGVs represent portosystemic collaterals, IGVs in sinistral portal hypertension are porto-portal collaterals that are supplied by the short and posterior gastric veins but drain back into the portal vein through the left and right gastric veins. It is important to differentiate sinistral portal hypertension from systemic portal hypertension because the pathophysiology and management of the two entities differ.

Rectal varices form at the junction of the superior rectal vein with middle and inferior rectal veins and differ from hemorrhoids because they do not extend to the dentate line. Bleeding from rectal varices and other ectopic varices is uncommon, and management should be considered on a case-by-case basis based on vascular anatomy.

TIPS

TIPS creation is a procedure in which a percutaneously created portosystemic shunt is used to treat the

TABLE 2 Classifications of gastric varices

	Basis of classification	Description
Sarin Classification ^[9]	Endoscopic location	Gastroesophageal varices (GOV) type 1: lesser curvature Gastroesophageal varices (GOV) type 2: greater curvature Isolated gastric varices type 1: fundus Isolated gastric varices type 2: anywhere in the stomach other than fundus
Saad-Caldwell Classification ^[4]	Type of varices and presence and absence of shunts	Type 1: Supply (afferent flow) through left gastric vein 1(a): Drainage through small portosystemic collaterals 1(b): Drainage mainly through GRS Type 2: Supply (afferent flow) through short gastric veins 2(a): Drainage through small portosystemic collaterals 2(b): Drainage mainly through GRS Type 3: Supply (afferent) through left gastric and short gastric veins 3(a): Drainage through small portosystemic collaterals 3(b): Drainage mainly through GRS Type 4 4(a): Type 3(a) with PVT 4(b): Type 3(b) with PVT
Kiyosue Classification ^[11]	Patterns of supplying (afferent) and draining (efferent) veins	Supply Type 1: Single gastric vein Type 2: Multiple gastric veins Type 3: Single or multiple gastric veins with coexistent gastric veins that are contiguous with a shunt but do not contribute to the varices Drainage Type A: Single shunt Type B: Single shunt and collateral veins Type C: Both gastorenal and gastrocaval shunts Type D: A shunt which cannot be catheterized
Hirota Classification ^[12]	Visualization during venography	Grade 1: No evidence of collateral veins, gastric varices are well opacified Grade 2: Few small collateral veins, contrast remains in gastric varices for > 3 min Grade 3: Few medium to large collateral veins, partial opacification of gastric varices, contrast remains in gastric varices <3 min Grade 4: Large collateral veins, gastric varices are not well opacified Grade 5: Large GRS with rapid flow; left adrenal vein cannot be occluded with balloon catheter
Matsumoto Classification ^[13]	Flow dynamics of the circuit	Type 1: Gastric veins communicate with GRS 1(a): Hepatopetal blood flow 1(b): Hepatofugal blood flow Type 2: No evidence of communication between gastric veins and GRS 2(a): Hepatopetal blood flow 2(b): Hepatofugal blood flow

complications of portal hypertension such as variceal hemorrhage, refractory ascites, and hydrothorax. Since its inception in the 1990s, the TIPS procedure has advanced in technique, applications, and devices (e.g., polytetrafluoroethylene [PTFE]–covered stent-grafts).

Physiological effects of TIPS

A TIPS is an endovascular shunt placed under radiographic guidance connecting the portal system with the systemic circulation and aiming to decrease the portal venous pressure gradient. Portal flow is diverted to the systemic circulation, resulting in an increase of about 50%–100% in right atrial pressure, cardiac preload, and central blood volume and about 10%–15% in cardiac output owing to improved cardiac inotropy.^[14,15] Meanwhile, systemic vascular resistance and, consequently, cardiac afterload decrease. Right ventricular pressure, pulmonary arterial pressure, and pulmonary capillary wedge pressure also increase after TIPS. Heart rate, although initially unchanged, may increase after several months. Plasma copeptin, aldosterone, and renin levels decrease, whereas norepinephrine levels increase after TIPS. Renal function may improve after TIPS through improved renal perfusion.^[14]

Technique

TIPS procedure

TIPS procedure is usually performed by interventional radiologists under real-time fluoroscopic and ultrasound (US) guidance. It is typically created using a transjugular venous approach under general anesthesia or deep sedation. A hepatic vein is catheterized, and, from within that vein, the portal vein is punctured using a long-curved needle. Once the portal vein is accessed, portal venography and hemodynamic assessments are performed, followed by dilation and measurement of the intrahepatic parenchymal tract. Finally, an expandable PTFE-covered stent-graft is used to line the tract, from portal vein entry to the hepatic vein ostium (Fig. 2). A shunt can be created between the IVC and portal vein as well, termed the direct intrahepatic portosystemic shunt (DIPS). DIPS is indicated when the hepatic veins are occluded or unsuitable, as can occur with Budd-Chiari syndrome. Bare metallic stents are no longer used in TIPS because of their higher rate of stenosis and occlusion compared with PTFE-covered stents, the use of which has also been shown to associate with decreased bleeding and improved survival.^[16]

Pre-TIPS workup should ideally include cross-sectional imaging to have an adequate anatomical map of the portal vein and hepatic veins. In emergent situations, bedside Doppler-US might be an alternative. An

echocardiogram should be performed to assess for the presence and severity of systolic and diastolic dysfunction and pulmonary hypertension. Recent AASLD guidance provides recommendations for preprocedural correction of coagulopathy.^[17]

The goal of portosystemic diversion is to optimize the balance between improvements in ascites, nutrition, renal function, and bleeding against the risk of worsening hepatic encephalopathy, cardiac overload, and liver function. Thus, creating the smallest-necessary caliber shunt to balance those outcomes is desirable. Recent studies suggest that an 8-mm-diameter PTFE-lined TIPS may be sufficient to prevent variceal rebleeding and potentially decrease the incidence of hepatic encephalopathy^[18–21] while lessening the possibility of worsened hepatic decompensation compared with larger shunts. In fact, the use of the 8-mm stent may lead to a survival advantage compared with those of the 10-mm stent.^[21,22] Additionally, the use of the 8-mm diameter stent has been found to be superior to medical therapy in the prevention of rebleeding from esophageal varices.^[19] Distinct from previously available stents, which had a fixed diameter, a more recently introduced TIPS stent-graft (so-called “controlled expansion”) allows a range of operator-determined shunt diameters to be created with a single shunt, between 8 and 10 mm.

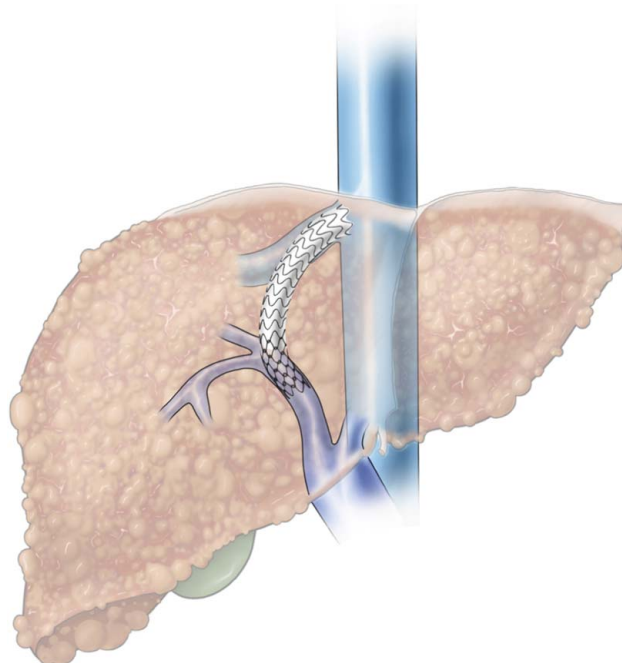


FIGURE 2 TIPS. The stent should cover the hepatic vein up to its ostium into the IVC to prevent dysfunction. The covered portion of the stent covers the parenchymal tract and the hepatic vein, whereas the uncovered portion of the stent is located in the portal vein.

Hemodynamic Pressure Measurements and Goals

The HVPG refers to the difference in intravascular pressure between the portal vein and the hepatic vein. During TIPS placement, direct portal pressures are measured and used to calculate the portosystemic pressure gradient (PSPG). As recommended by two recent consensus conferences,^[23,24] PSPG should be calculated by subtracting the suprahepatic (intra-abdominal) IVC pressure from portal pressure to better correlate with clinical outcomes.^[25] When the IVC pressure cannot be measured because of technical difficulty, the right atrial pressure is often used in lieu of IVC pressure, but this may lead to an overestimation of the PSPG because atrial pressure is generally lower than IVC pressure.^[23,24] Additionally, portal and systemic hemodynamics recorded during general anesthesia or conscious sedation, which are generally done in fasting conditions, may not reflect the “real” PSPG.^[26]

Post-TIPS PSPG level is predictive of rebleeding risk because lower values are associated with lower bleeding recurrence and, inversely, predictive of post-TIPS complications such as hepatic encephalopathy and worsened liver function (lower values are associated with higher rates of complications).^[27] Thus, the PSPG decrease target must balance bleeding control with safety in an individualized approach by indication and patient profiles.

In patients with acute, uncontrolled esophageal variceal bleeding, the desired post-TIPS PSPG is <12 mm Hg^[26,27] or a reduction $\geq 50\%$ from baseline PSPG.^[28] However, the ideal post-TIPS PSPG for the management of gastric variceal bleeding is not well defined. The desired PSPG for secondary prevention of gastroesophageal variceal bleeding is <12 mm Hg. In a recent study, patients with a PSPG ≥ 12 mm Hg after TIPS had a 35% risk of recurrence of portal hypertension complications at 2 years, as compared with <5% in those with a PSPG <12 mm Hg.^[26] A PSPG > 12 mm Hg at any time during follow-up was associated with 8.5-fold increase in the rate of recurrence of portal hypertension complications (bleeding and/or ascites).^[26] In patients not achieving a PSPG <12 mm Hg despite dilation of the stent to a maximum 10 mm of diameter, the addition of nonselective beta-blockers (NSBBs) should be considered because they might reduce portal pressure by an additional 20%–25%.^[29,30]

Presence of PVT

Acute or chronic bland (noncancerous) PVT are not a contraindication to TIPS, although the complexity of shunt creation and thrombectomy may require specialized technical skills. Various techniques such as transhepatic and trans-splenic approaches have proven useful in

facilitating shunt creation, venous thrombectomy, and vascular recanalization.^[31,32] The success of TIPS in PVT is high, with 84% 1-year TIPS stent patency rate and 78% 1-year portal vein patency rate, although the incidence of major complications is 10%.^[33,34] TIPS failure and shunt dysfunction in the setting of PVT are affected by the degree of portal vein and superior mesenteric vein thrombosis and the presence of underlying thrombophilia.^[35] Portal vein recanalization and TIPS may also improve a patient's candidacy for liver transplantation^[36] by restoring patency to the portal vein, making it suitable for surgical anastomosis and graft perfusion.

Variceal Embolization

Embolization of gastric, esophageal, or ectopic varices may be performed during TIPS creation (see also the section on 4.2 on ATO). The indications for embolization include actively bleeding varices or persistent fugal variceal filling. In addition, embolization of enlarged portosystemic shunts (with or without varices) may be considered to prevent severe hepatic encephalopathy in patients receiving TIPS.^[37] Rebleeding may decrease with adjunctive antegrade embolization of residual varices in patients treated with TIPS for bleeding indications,^[38] but the results of two RCTs have been inconsistent.^[39,40] Ectopic and gastroduodenal varices typically require embolization independent of shunt creation because downhill portosystemic flow through the varix may be incompletely diverted at target PSPG levels suitable for esophageal varices.

Guidance statements

1. PTFE-coated TIPS stents should be considered standard of care.
2. When the indication for TIPS is variceal hemorrhage (treatment of acute hemorrhage or prevention of recurrence), TIPS should be progressively dilated (starting at 8 mm of diameter) to the minimum diameter needed to achieve a PSPG below 12 mm Hg.
3. In patients whose PSPG does not decrease below 12 mm Hg despite maximum dilation of TIPS (10 mm), NSBBs should be added to further decrease portal pressure.
4. Bland PVT does not preclude creation of a TIPS. Referral to experienced centers should be considered.
5. In patients with large spontaneous portal systemic collaterals, collateral embolization at the time of TIPS placement may be considered

TABLE 3 Summary of endovascular therapies in the management of variceal hemorrhage

	Variceal obliteration techniques				
	TIPS	BRTO	RTO PARTO	CARTO	ATO
Technique	Percutaneously created portosystemic shunt connecting portal vein with hepatic vein TIPS can be performed with ATO. Might complement RTO	Balloon occlusion of a GRS and retrograde injection of sclerosant agent, with ensuing gastric variceal obliteration Balloon occlusion maintained up to 36 h Need for ICU monitoring	The GRS is occluded with a vascular plug, and Gelfoam is injected retrogradely to obliterate the gastric varices.	The GRS is occluded with coils, and Gelfoam is injected retrogradely to obliterate the gastric varices.	Percutaneous transhepatic or trans-splenic embolization of portomesenteric veins supplying esophageal, gastric or ectopic varices using various embolic materials ATO can be performed with TIPS or RTO.
Technical limitations	Challenging in patients with PVT, biliary dilation, hepatic tumors, and polycystic liver	Need for prolonged balloon occlusion (up to 36 h), with ICU monitoring during this period Might not be possible in patients with shunts larger than available balloon diameters	Might not be possible in patients with shunts larger than available plugs (< 18 mm) Might not be possible in challenging anatomy, such as a very tortuous course from IVC to renal vein to GRS (Z-path)	Might not be possible in patients with giant shunts larger than available coils (< 30 mm) No anatomical limitation	Might not be able to embolize multiple collateral veins supplying varices, resulting in early rebleeding from collaterals More invasive than RTO because it requires direct percutaneous access through the liver or spleen or through TIPS
Physiological effects	Decrease in PSPG Increase in CO Increase in cardiopulmonary pressures	Decrease in portosystemic shunting Redirection of flow through portosystemic collaterals to the liver, with potential improvement in liver function and hepatic encephalopathy Increase in PSPG with the possibility of new complications of portal hypertension (ascites or esophageal variceal hemorrhage)			
Clinical context of use	Variceal hemorrhage because of esophageal and GOV1, gastrofundal varices or ectopic varices Preemptive TIPS Salvage TIPS Rescue TIPS Secondary prevention	Gastrofundal variceal hemorrhage with a GRS allowing retrograde access: Treatment of acute hemorrhage Prevention of rebleeding Treatment of ectopic varices with favorable anatomy (large and accessible efferent shunt)			Variceal hemorrhage because of Esophageal varices Gastric varices Ectopic varices
Contraindications	Congestive heart failure (Stage C/D or EF <50%) Severe pulmonary hypertension with mPAP > 45 mm Hg Severe uncontrolled hepatic encephalopathy Uncontrolled sepsis	Splenic and/or portal vein thrombosis (needs sufficient hepatopedal flow capacity for blood flow redirection) Active sepsis Uncontrolled esophageal variceal hemorrhage (would require simultaneous TIPS) Poor liver function or hepatic encephalopathy are not a limitation and might improve with RTO			Coagulopathy Splenic and/or portal vein thrombosis (unable to access these vessels) Active sepsis
Procedural or short-term complications	< 5% frequency Intraperitoneal bleeding Arterial injury Liver infarct Capsular puncture Hemobilia	Related to sclerosant escape to systemic and/or portal circulation: Renal vein thrombosis PVT Pulmonary embolism Anaphylactic shock	Related to Gelfoam escape to systemic and/or portal circulation (rare) Renal vein thrombosis PVT Pulmonary embolism Plug or coil migration into renal vein and causing nontarget embolization (rare)	Related to embolic materials escape to systemic and/or portal circulation (rare) Renal vein thrombosis PVT Pulmonary embolism	

Long-term complications	Sepsis	Disseminated intravascular coagulation Acute kidney injury	Related to the increase in PSPG New or worsening esophageal or ectopic varices New or worsening ascites and/or hepatic hydrothorax
	TIPS thrombosis/dysfunction		
Follow-up	Related to an increase in portal systemic shunt: New or worsening hepatic encephalopathy Worsening liver function Related to cardiac overload Congestive heart failure Pulmonary hypertension		
	Doppler-US study in 1–4 weeks 3 months 6 months and 6 months thereafter	Contrast CT venogram or EUS in 2–3 days prior to discharge Endoscopic evaluation of gastric and esophageal varices in 1–2 months If normal, further imaging and endoscopic follow-up according to existing guidance	

Abbreviations: CO, cardiac output; EF, ejection fraction; EUS, endoscopic ultrasound; GOV1, gastroesophageal varices type 1; GRS, gastroduodenal shunt; mPAP, mean pulmonary artery pressure; PVT, portal vein thrombosis.

because it may decrease the risk of hepatic encephalopathy.

Contraindications

Absolute contraindications for TIPS include congestive heart failure (stage C or D, or a documented ejection fraction <50%), severe pulmonary arterial hypertension (mean pulmonary artery pressure of > 45 mm Hg), severe uncontrolled hepatic encephalopathy, and systemic infection or sepsis^[41] (Table 3). Untreated biliary obstruction and uncorrectable severe coagulopathy are relative contraindications. Creating a TIPS may be difficult in patients with polycystic liver disease, hepatic or portal vein occlusion, and intrahepatic tumors, although successful TIPS creation has been reported in these settings. Although higher model for end-stage liver disease (MELD) score is associated with higher mortality,^[42,43] no specific MELD threshold can be recommended to contraindicate TIPS. Most trials assessing the use of TIPS for variceal bleeding excluded patients with a Child-Pugh score > 13 points. Patients over 75 years old have been also excluded from RCTs, and performing TIPS in patients over 70 has been infrequent.^[44] A recent study suggests acceptable outcomes of TIPS in highly selected patients over 70 years old.^[44]

It is important to note that risk-benefit tradeoffs are different when TIPS is a last resort, life-saving procedure (e.g., uncontrolled variceal hemorrhage) versus when alternative treatments exist (e.g., secondary prevention of variceal hemorrhage or refractory ascites). Futility criteria for TIPS placement in the context of variceal hemorrhage are discussed later in this document.

Complications

Procedural complications during TIPS are rare (<5%) in contemporary series^[45,46] but may include intra-peritoneal bleeding, arterial injury, liver infarct, hepatic capsular puncture, hemobilia, immediate TIPS thrombosis, and sepsis.^[47,48] Procedure-related deaths occur in <1%.^[46]

The most relevant long-term complications are those related to shunting, mainly hepatic encephalopathy, deterioration in liver function, and complications related to cardiac overload (Table 3).

The probability of developing at least one episode of hepatic encephalopathy ranges between 30% and 50%^[49–51] and that of severe, disabling encephalopathy is approximately 8%.^[50] Risk factors for hepatic

encephalopathy are previous hepatic encephalopathy, older age, advanced liver dysfunction, kidney dysfunction, hyponatremia, sarcopenia, and a post-TIPS low PSPG.^[52] As mentioned previously, 8-mm TIPS stents (which are associated with higher post-TIPS PSPG) are associated with lower incidence of hepatic encephalopathy compared with that of 10-mm TIPS stents while maintaining similar effectiveness against variceal rebleeding and TIPS patency.^[19] In general, post-TIPS encephalopathy can be managed with medical therapy using lactulose and rifaximin in the vast majority of cases. In a recent double-blind multicenter RCT, prophylactic rifaximin, starting 2 weeks before TIPS placement, reduced the risk of overt hepatic encephalopathy.^[49] In 81% of the patients included in the trial, TIPS was performed to treat ascites, and the etiology of cirrhosis was alcohol-related in 86% of the cases. This leaves some uncertainty regarding the extrapolation of these results to patients with a bleeding indication for TIPS and with causes of liver disease other than alcohol. If encephalopathy is refractory to medical therapy, endovascular reduction techniques can be used to reduce TIPS diameter (TIPS recalibration).^[53] If the PSPG after TIPS reduction is greater than 12 mm Hg, the patient should be treated with NSBBs or undergo screening/surveillance endoscopy with endoscopic treatment of varices as appropriate.

Portosystemic shunting might lead to a deterioration in liver function, but the incidence of this complication is difficult to estimate in the context of the natural progression of the underlying liver disease. TIPS induces a slight increase in bilirubin and international normalized ratio,^[54] without a signal for increased liver failure-related deaths.^[51,54,55]

A recent prospective study showed a rate of cardiac decompensation of 20% after TIPS,^[56] but half of the patients experiencing decompensation had a previously diagnosed cardiomyopathy. Post-TIPS mortality related to heart decompensation was 5% within 1 year, which emphasizes the relevance of pre-TIPS cardiac evaluation. In a retrospective series, TIPS was associated with a new development of pulmonary hypertension in 4% of the patients.^[57]

Follow-up after TIPS placement

TIPS dysfunction, defined as the loss of portal venous decompression (resulting in PSPG > 12 mm Hg), can be due to TIPS thrombosis or stenosis. TIPS dysfunction manifests clinically with recurrence of portal hypertension-related symptoms (recurrent variceal bleeding, ascites, or hepatic hydrothorax). The incidence of TIPS dysfunction varies and is far more likely with bare metallic stents.^[58] With PTFE stent-grafts, the most common site of loss of patency and recurrent

portal hypertension remains the unstented native hepatic vein, emphasizing the need for precise shunt placement to the vein ostium. The patency rates of PTFE stents are above 90% at 1 year and 75% at 3 years.^[59]

In patients who underwent TIPS placement for prevention or treatment of variceal hemorrhage, a proactive follow-up plan with Doppler-US of the TIPS is recommended to detect subclinical TIPS dysfunction. This might require referral to a tertiary center because TIPS interrogation with Doppler-US is not a routine procedure. With covered stents, initial Doppler-US evaluation is usually performed 1–4 weeks, 3 months, and 6 months after TIPS creation and every 6 months thereafter.^[60] The combined use of mean maximum flow velocity (mVPmax) at the portal vein and direction of intrahepatic portal flow predict TIPS dysfunction with high sensitivity and good negative likelihood ratio, thus avoiding the need for TIPS interrogation in equivocal cases.^[61] The prediction rule for TIPS dysfunction derived from this model (mVPmax <28 cm/s when flow is hepatofugal or mVPmax <39 cm/s when flow is hepatopetal) had 90% sensitivity, 45% specificity, and negative likelihood ratio of 0.23.^[61] The accuracy of Doppler-US for detecting TIPS dysfunction can be improved if clinical parameters (i.e., presence of ascites) are taken into consideration along with the Doppler-US findings.^[62] Endovascular evaluation and therapy are recommended when Doppler-US demonstrates TIPS stenosis or occlusion. On the other hand, despite the high negative predictive rate of Doppler-US, a “negative” Doppler-US in the setting of recurrent portal hypertension signs or symptoms should not necessarily preclude direct shunt interrogation by catheter venography.

Guidance statements

6. In patients undergoing elective TIPS for prevention of variceal rebleeding, rifaximin 550 mg BID started within 14 days before TIPS placement and maintained for 6 months may reduce the risk of post-TIPS hepatic encephalopathy and can be considered when feasible.
7. In patients with refractory hepatic encephalopathy despite post-TIPS optimal medical therapy (combination of lactulose and rifaximin), endovascular reduction of TIPS diameter should be attempted.
8. In TIPS placed for variceal hemorrhage, we suggest a frequency of follow-up Doppler-US

- at 1 week, 3 months, 6 months, and every 6 months thereafter to assess TIPS patency.
9. A mean maximum flow velocity (mVPmax) at the portal vein <28 cm/s when flow is hepatofugal or mVPmax <39 cm/s when flow is hepatopetal should trigger venography and pressure measurements to confirm TIPS dysfunction. An absence of these positive findings should not preclude TIPS interrogation in patients with recurrent signs or symptoms of portal hypertension.

VARICEAL OBLITERATION TECHNIQUES: RTO AND ATO

Endovascular variceal obliteration or embolization, RTO or ATO, are alternative or complementary approaches to TIPS for variceal bleeding (Table 3). These act directly on varices and do not decrease portal pressure. In fact, obliterating large collaterals can actually increase portal pressure and consequently increase PSPG. Variceal obliteration can be used in combination with TIPS when persistent varices are seen after TIPS placement or as a first-line endovascular treatment (without TIPS) for patients with refractory variceal bleeding who are not good candidates for TIPS placement, such as those with poor hepatic reserve, severe pulmonary hypertension, or poorly controlled hepatic encephalopathy. RTO is mainly performed for gastric or ectopic varices in which the varices are accessed from the systemic veins, as in the case of a gastroduodenal or gastroduodenal shunt. ATO can be used to treat esophageal (mostly in combination with TIPS), gastric, and ectopic varices and generally requires access through a percutaneous transhepatic or transsplenic approach or through TIPS.

RTO

Although the first RTO was reported in 1984 by Olson et al in the United States,^[63] it was mostly performed and further developed in Japan and Korea, and the first balloon-occluded RTO (BRTO) was reported in 1996 by Kanagawa et al.^[64] In the early 2000s, many U.S. institutions began performing BRTOs, and the first reported BRTO in the literature was published by Sabri et al in 2011.^[65] Shortly after, the plug-assisted RTO (PARTO) and coil-assisted RTO (CARTO) were invented and published in 2013 by Gwon et al^[66] and in 2014 by Lee et al,^[67] respectively. These modifications, described as follows, substantially simplified the

procedure and reduced complications, leading to a wider acceptance of RTO as a treatment option for gastric variceal bleeding.

Rational basis and physiological effects of RTOs

The primary aim of all RTO procedures is to completely obliterate varices that have recently bled or are actively bleeding. With RTO, resolution of variceal hemorrhage from gastroduodenal varices occurs in 96%–100% with a recurrence rate of <3% up to 3–10 year follow-up.^[68] RTOs have a high technical success rate of 97%–100% with <3% major complication rates. However, a prerequisite for performing RTO is the presence of adequate antegrade portal venous flow, either through a patent portal vein or through porto-portal collaterals (cavernous transformation), because obliteration of the variceal complex (efferent veins, varices, and afferent vein/shunts) leads to restoration of blood flow through the portal vein into the liver.^[69] Additionally, because RTO does not have any effect on portal venous resistance, the redirection of flow to the liver may increase portal pressure and PSPG,^[70] which can lead to the development of new collateral pathways to the often coexisting esophageal varices in up to 60% of cases.^[50,71,72] Ascites and hepatic hydrothorax may also worsen in up to 10% after the procedure, which may necessitate subsequent TIPS placement.^[73] As a result, patients with esophageal bleeding and/or refractory ascites/hydrothorax following RTOs may require treatment with TIPS placement.^[74] On the other hand, the increased portal flow following RTOs may help improve hepatic function, leading to improved MELD and Child-Pugh scores.

Technique

Prior to RTO, comprehensive imaging evaluation including computed tomography (CT) or magnetic resonance (MR) venogram with contrast and coronal reconstruction images is strongly recommended to understand and anatomically map the afferent veins, gastric varices, efferent veins/shunt (GRS), collaterals, renal vein, and IVC. The RTO procedure can usually be performed under moderate sedation, but monitored anesthesia care or general anesthesia may be needed in patients who are more critical.

Traditional BRTO

In a traditional BRTO procedure (Fig. 3), a balloon catheter is inflated in the GRS as a protectant measure (to prevent escape of sclerosant to systemic circulation), and then a sclerosing agent (typically a sodium tetradecyl

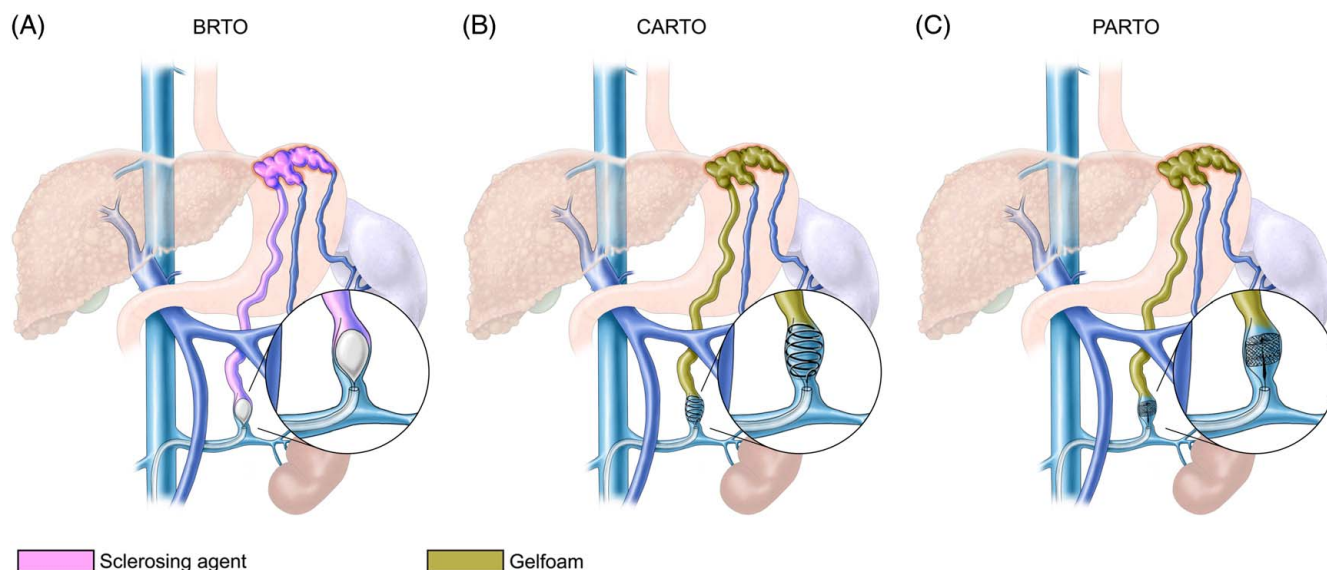


FIGURE 3 Diagrams of retrograde transvenous obliteration (RTO) procedures. (A) BRTO with a balloon inflated in the GRS and a sclerosing agent (sodium tetradecyl sulfate) injected retrograde into the gastrofundal varices and gastroduodenal shunt. (B) PARTO with a vascular plug is used to occlude the gastroduodenal shunt, and Gelfoam is filling the variceal complex. (C) CARTO with coils deployed and occluded the gastroduodenal shunt, and Gelfoam is filling distally to obliterate the variceal complex.

sulfate solution in the U.S. or ethanolamine oleate in Asia) is injected retrogradely into the GRS and gastrofundal varices through the balloon occlusion catheter. The balloon occlusion catheter must remain inflated for an extended period of time, up to 36 h, to obliterate gastrofundal varices and to protect the renal and portal veins.^[75] During this time, the patient is required to be monitored in the intensive care unit (ICU) because the indwelling balloon has the potential to rupture, which can lead to catastrophic complications. Repeated treatments may be required to obliterate all the existing collateral networks and varices.

Modified BRTO

With the risk of balloon rupture, complications associated with sclerosing agents, ICU admission, multiple interventional radiology procedures, and longer procedure time as the negative hallmarks of the BRTO procedure,^[76] the modified BRTO (mBRTO) procedure has been developed and has been shown to lower the risk of such complications. The mBRTO procedure uses balloon occlusion with additional embolic devices, vascular plugs, or coils to improve the protection from balloon rupture at the GRS.^[66,67] Nevertheless, mBRTOs still use sclerosing agents, which are potentially toxic and responsible for many of the complications and adverse outcomes associated with BRTOs.^[67]

PARTO

The PARTO procedure is an alternative to the BRTO or mBRTO that eliminates the use of both the balloon

occlusion catheter and the sclerosing agent. Instead, a vascular plug (typically the Amplatzer plug) and Gelfoam slurry are used to obliterate the gastric varices.^[77] The use of Gelfoam instead of sclerosing agents obviates the need for embolization of small collateral vessels, which is an essential part of BRTOs to prevent leakage of sclerosing agents.^[73] The PARTO procedure is performed by accessing the GRS with a catheter, which is then upsized to a large sheath to deploy the vascular plug within the most narrow portion of the shunt (choke point). After plug deployment, additional embolization of the entire gastric variceal complex is performed using Gelfoam. However, there are several technical considerations involved in the PARTO procedure that may make it challenging, such as shunt size and anatomy.^[78] For example, a patient with a large shunt may not be a candidate for PARTO because there is no vascular plug large enough to occlude the shunt.

CARTO

The CARTO procedure is an alternative to the BRTO, mBRTO, and PARTO procedures.^[75] Instead of the balloon and sclerosing agents in BRTO or the vascular plug in PARTO, the CARTO uses a combination of endovascular coils and Gelfoam to obliterate gastroduodenal varices. Instead of a balloon catheter, the GRS (efferent shunt) is accessed with two catheters. The first or distal catheter is advanced distally into or adjacent to the gastrofundal varices, and the second or proximal catheter is placed proximally within the GRS near the renal vein to deploy correctly sized detachable coils to completely occlude the shunt. After coil embolization of the GRS by the proximal catheter, the distal catheter is used to perform Gelfoam embolization of the entire

gastric variceal complex. Like the PARTO technique, CARTO also aims to obliterate the gastric varices in a single session without the use of any toxic sclerosing agents or prolonged inflation of the balloon catheter. Unlike PARTO, there is no shunt sizing restriction, anatomical challenges, or dependence on the availability of balloons or plugs with CARTO. Therefore, preprocedure planning is simplified with CARTO, and a larger GRS can be treated because the coils can be appropriately sized to any blood vessels.

Contraindications

Relative contraindications of RTOs include severe sepsis, splenic or portal vein thrombosis, and uncontrollable esophageal variceal bleeding.^[71] High MELD or Child-Turcotte-Pugh (CTP) scores are not contraindications for RTOs, in contrast to TIPS. The presence of difficult-to-control ascites is a relative contraindication that can worsen with RTO and can be mitigated by TIPS placement after RTO.^[71,73] In addition, several anatomical factors must be considered. There must be an efferent shunt (e.g., GRS) amenable to catheterization and occlusion. PARTO procedures can be sometimes challenging because of the anatomy of a zigzag pathway (Z-pathway) from IVC-renal vein-GRS, which may hinder deployment of the plug (Fig. 4). Lastly, the GRS size can also limit the BRTO and PARTO procedures because the balloon occlusion catheter or vascular plug, respectively, may be too small to occlude the shunt.

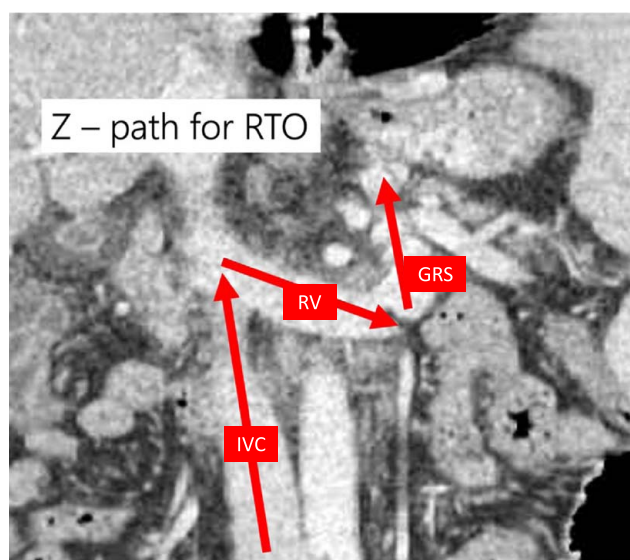


FIGURE 4 Illustration of the “z-path” involving IVC, renal vein (RV), and gastorenal shunt (GRS).

Complications of RTOs

Traditional BRTO procedures require potentially toxic sclerosing agents such as sodium tetradecyl sulfate solution or ethanolamine oleate. If these agents escape into the systemic or portal circulation, they can cause portal and/or renal vein thrombosis^[71] and catastrophic complications including anaphylactic shock, stroke, and disseminated intravascular coagulation.^[67,78] Additional complications, such as gross hematuria, cardiac arrhythmia, pulmonary embolization, and renal failure, are rare but have been reported with these agents.^[71]

Other complications of RTO procedures are related to increased portal pressure and increased PSPG caused by RTOs. These include new or worsening esophageal or ectopic varices with variceal bleeding in up to 33%^[69,71,75] as well as worsening of ascites and/or hepatic hydrothorax.^[74]

Follow-up care after RTO procedures

Complete obliteration of the variceal complex (afferent vein, gastric varices, and efferent vein) following RTOs is crucial because partial obliteration or embolization may cause massive bleeding because the only available decompressing shunt is occluded while the varices are still present. Therefore, a contrast-enhanced CT of the abdomen should be obtained within 2–3 days of the RTO to confirm complete obliteration of the variceal complex. Where available, endoscopic US can also be used for this purpose. If complete obliteration is not achieved, the patient may require additional procedures such as ATO, TIPS, or endoscopic therapy on an individualized basis. Further imaging should be guided by other clinical needs, including hepatocellular carcinoma screening and surveillance.

With a successful RTO, given the risk of increased PSPG, routine endoscopic evaluation to assess for new development or progression of esophageal varices after 1–2 months is recommended. Further treatment of esophageal varices should follow current applicable Guidelines.^[1] Finally, because RTO has the potential to increase portal pressure with ensuing complications, treatment with NSBBs could be considered in patients with high-risk esophageal varices without contraindications.

Guidance statements

10. RTO should be considered as an alternative treatment to TIPS for bleeding gastric or ectopic varices with favorable anatomy (e.g., accessible and occludable GRS).
11. RTO, by closing the flow through the varices, redirects portal venous flow to the liver and thereby increases the portal

systemic pressure gradient. Close surveillance of complications of portal hypertension and potential additional treatment such as TIPS, beta-blockers, or endoscopic treatment of esophageal varices after RTOs is needed.

12. CARTO or PARTO have lower complication profiles compared with those of traditional BRTO or mBRTO and may be the preferred methods at experienced institutions.
13. Within 72 h after RTO, a CT of the abdomen or endoscopic US should be performed to confirm complete obliteration of the gastric or ectopic varices.
14. Follow-up endoscopy should be performed 1–2 months after RTO to survey for the development or progression of esophageal varices. If no varices are seen, future surveillance should be performed according to existing guidance.

because TIPS provides easy access to the afferent vessels of varices.

Physiological effects

Similar to RTO, ATO redirects collateral flow to the liver, thereby increasing portal pressure and PSPG, which may improve liver function and hepatic encephalopathy and may cause a progression of other complications of portal hypertension such as variceal bleeding and ascites/hydrothorax. The magnitude of the increase in portal pressure depends on the size (and flow) of the occluded collateral(s) and should be taken into account in determining whether ATO itself can be a definite treatment or if it should be followed by TIPS placement to control portal pressure. Technical success of ATO is defined as the complete obliteration of the varices and cessation of flow after completion of the embolization and varies between 44% and 100%.^[80] This wide range of success rate is because, frequently, multiple afferent or feeding collaterals can supply varices, and some of the feeding collaterals could be missed during the ATO procedure.

ATO

The first ATO of gastroesophageal varices was performed by Lunderquist and Vang in 1974.^[79] The procedure is performed to decrease venous inflow and to obliterate the varices through embolization. ATO has been used to treat esophageal, gastric, and ectopic varices, generally in the setting of a large afferent vein or shunt or with high flow varices. Specifically, when no GRS is present in patients with gastrofundal varices, RTO cannot be performed and ATO may be chosen. Lastly, ATO is often performed in conjunction with TIPS

Technique

The ATO procedure can be performed under moderate sedation, monitored anesthesia care, or general anesthesia. CT or MR imaging helps delineate the anatomy of the varices, their feeding vessels, and the draining veins. ATO can be performed through a percutaneous transhepatic or trans-splenic access or a trans-TIPS access. A percutaneous access is performed through the portal or splenic vein, followed by a venography to delineate the portal venous anatomy to identify the branches feeding the varices, and then embolization is performed using a variety of embolic materials, such as

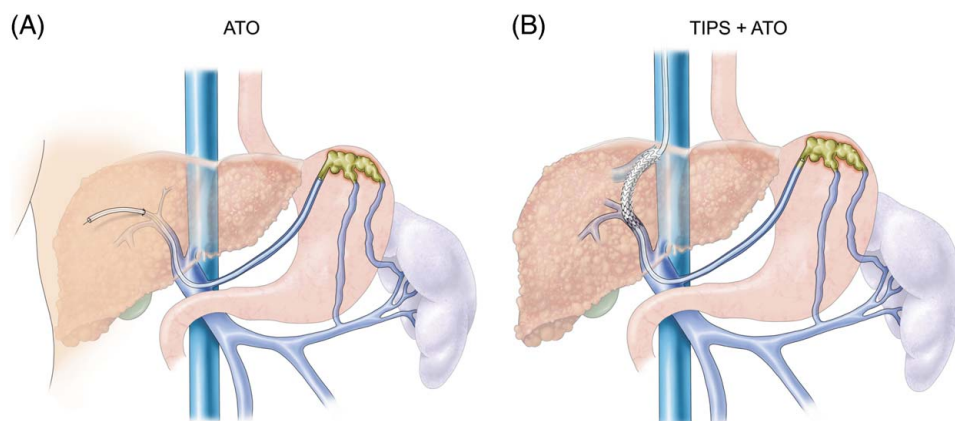


FIGURE 5 ATO performed for the obliteration of gastric varices. ATO is performed either (A) through percutaneous access into portal venous system or (B) through a TIPS. The three main inflow branches of the portal vein or splenic vein that supply gastric varices are left gastric vein (the “coronary vein”), posterior splenic vein, or short gastric veins. In the case in the figure, the main feeder of the variceal nidus is the left gastric vein. The inflow branches are then catheterized individually, and embolization of the perigastric inflow veins and gastric varices is performed.

sclerosing agents, Gelfoam, n-butyl cyanoacrylate (glue), coils, or plugs.^[81–85] Once embolization is completed, the percutaneous access tract should also be embolized for hemostasis. Figure 5 shows the procedure as it is performed for gastric varices. ATO can be performed through the TIPS access, as detailed in later sections (see section 5.2.2.1).

As with RTO, complete obliteration of varices is also absolutely crucial with ATO because incomplete obliteration has been associated with a high rebleeding rate because of regrowth of residual varices with collateral supplies.^[70,83,84,86,87] Therefore, the goal of the embolization should be complete embolization of the afferent veins as well as complete obliteration of the varices without any nontarget embolization into systemic or portal circulation.

Because it is less invasive procedure, RTO is preferable to ATO. However, ATO can be performed as an adjunctive procedure to RTO, when RTO fails, or if RTO is not technically feasible. Combining ATO with RTO can increase the success rate of variceal obliteration and can decrease the chance of over-spillage of embolic materials into the portal system.^[80]

Contraindications

Coagulopathy is a relative contraindication when ATO is performed through a percutaneous access because it may increase periprocedural access site hemorrhage. We refer to the specific AASLD Guidelines for the assessment and management of procedural bleeding risk.^[17] In patients with splenic and/or portal vein thrombosis, it might not be possible to access the collaterals feeding the varices. Uncontrolled sepsis is an absolute contraindication.

Complications

The main complications reported after ATO are fever and abdominal pain, which are usually transient, and those related to increased portal pressure, such as worsened esophageal varices and new or increased ascites and/or hydrothorax.^[86,88] Other less frequent complications include bleeding from access sites, hemoperitoneum, PVT, transient bacteremia, nontarget embolization, and renal vein thrombosis.^[86,88]

Follow-up after ATO procedure

A similar approach as used with RTO to confirm obliteration of varices and subsequent follow-up should be used, including a CT scan of the abdomen, and endoscopic follow-up to assess for new gastroesophageal varices.

Guidance statements

15. Percutaneous transhepatic or trans-splenic access or trans-TIPS ATO should be considered as a therapeutic option in esophageal, gastric, or ectopic variceal hemorrhage with a large afferent vein or shunt or with high flow varices.
16. ATO with complete eradication of entire variceal complex is advisable to minimize recurrent bleeding. The choice of embolic or sclerosing agent should be determined by operator experience and agent availability.
17. Abdominal CT or endoscopic US should be performed within 72 h after ATO to confirm complete obliteration of the varices.
18. Follow-up endoscopy should be performed 1–2 months after ATO to evaluate the development or worsening of esophageal varices. If no varices are seen, future surveillance should be performed according to existing guidance.

CLINICAL USE OF ENDOVASCULAR THERAPIES IN THE MANAGEMENT OF VARICEAL BLEEDING

The role of TIPS in the management of esophageal varices and gastroesophageal varices type 1 (GOV1)

This section will focus on the role of TIPS in the treatment of esophageal and gastroesophageal varices type 1 (GOV1), in which the management is considered similar to the management of esophageal varices. Because of the anatomy of these varices, RTO procedures are generally not indicated, so the discussion will focus on the role of TIPS.

Primary prevention of esophageal variceal bleeding

TIPS has no role in the primary prevention of variceal hemorrhage. Shunts should not be created to prevent first variceal hemorrhage, independent of the variceal size or characteristics. This is particularly important in patients with compensated cirrhosis in whom TIPS could lead to liver dysfunction and decompensation by diverting blood flow away from the liver.

Management of acute variceal bleeding

Role of TIPS as preemptive therapy

Current standard of care for patients with acute variceal bleeding consists of a combination of restrictive transfusion strategy, vasoactive drugs, prophylactic antibiotics, and endoscopic variceal ligation.^[8,23,89] However, up to 10%–20% of patients rebleed within 5 days despite current standard of care therapies, and those at high risk for rebleeding can be considered for preemptive TIPS placement. When preemptive TIPS is considered, it should be placed shortly after bleeding is controlled (within 72 h of admission upper endoscopy; thus, the term “early” TIPS has also been used).

Unfortunately, most of these patients also have Child-Pugh C cirrhosis,^[90,91] so TIPS can lead to liver failure (acute-on-chronic) with 6-week mortality rates ranging from 27% to 55%.^[92–94] Therefore, the risks versus benefits of preemptive TIPS need to be considered carefully, and the key is the selection of patients at high risk who would benefit from preemptive TIPS. The efficacy of preemptive TIPS was first reported in a controlled trial by Monescillo et al^[95] wherein 52 patients with an HVPG > 20 mm Hg were considered at high risk for failure^[90,96] and randomized to undergo preemptive TIPS versus standard of care therapies. In these patients, TIPS placement within 24 h of admission led to significantly lower rebleeding and mortality compared with standard of care. The results of this pioneer trial, however, have not been widely applied because the current standard of care of acute variceal bleeding at most centers does not include routine measurement of HVPG. A second RCT of 63 patients by García-Pagán et al^[51] addressed a similar question but defined patients at high risk based on clinical factors on admission, namely Child C (notably, those with CTP scores > 13 were excluded) and Child B with active bleeding at time of initial endoscopy despite treatment with vasoactive drugs. Other groups of patients who were excluded were those with isolated gastric varices, PVT, HCC, renal failure, or prehepatic portal hypertension and those older than 75. In this trial, preemptive TIPS placed within 72 h of initial upper endoscopy improved survival as compared with standard therapy (combination of endoscopic therapy with NSBBs). These results were later confirmed in a larger trial of 132

patients.^[55] Subsequently, an individual patient data meta-analysis, including 3 RCTs and 5 observational studies inclusive of 1,327 patients (310 received preemptive TIPS and 1,017 received drugs plus endoscopy; 224 patients from RCTs and 1,103 patients from observational studies) and inverse probability of the treatment weight approach to balance the 2 treatment groups,^[97] concluded that patients with Child C with CTP scores of 10–13 and patients with Child B with a CTP score of 8–9 who had active variceal bleeding at the time of initial endoscopy were patients at high risk who would likely benefit from preemptive TIPS performed early after initial endoscopic and pharmacologic therapy for acute variceal bleeding.

However, a more recent RCT of 58 patients published after the aforementioned meta-analysis in 2021 found similar 1-year survival between the standard of care and the preemptive TIPS groups (76.9% vs. 75.9%, $P=0.91$),^[98] and an updated meta-analysis inclusive of this newer trial did not find significant survival benefit with preemptive TIPS, although the 6-week rebleeding risk was lower with preemptive TIPS.^[99]

Both the recent ALTA^[24] and Baveno consensus conference reports^[23] recommend placement of PTFE-covered stents within 72 h in patients bleeding from EV, GOV1, and GOV2 who meet the criteria established in the individual patient meta-analysis (Child-Pugh class C 10–13 points or Child-Pugh class B 8–9 points with active bleeding at initial endoscopy). Refinement of the candidate population may occur once results of subsequent larger trials on preemptive TIPS become available because the data from the individual patient meta-analysis mentioned previously were still largely driven by observational data.^[100]

Data from observational studies suggest that the presence of encephalopathy, hyperbilirubinemia, and/or acute-on-chronic liver failure (ACLF) at the time of bleeding had no impact on survival in patients treated with preemptive TIPS^[101] and should not contraindicate preemptive TIPS, as per the Baveno consensus.^[23] However, because patients who were included in these observational studies likely underwent careful selection, these recommendations require further confirmation from additional studies and real-world experience. Additionally, randomized trials of preemptive TIPS to date have included patients with two of the most potentially reversible etiologies of liver disease,

TABLE 4 Clinical scenarios in which TIPS should be considered for the treatment of variceal bleeding

Preemptive TIPS (previously called “early TIPS”)	TIPS placed preemptively within 72 h of the hemorrhage, in patients with hemorrhage controlled with medical/endoscopic therapy at high risk of treatment failure
Salvage TIPS	TIPS placed in patients with uncontrolled (ongoing) bleeding despite medical and endoscopic therapy
Rescue TIPS	TIPS placed for early (within 5 days) recurrence of variceal hemorrhage
TIPS for secondary prevention of variceal hemorrhage (“elective TIPS”)	TIPS placed in patients who have recovered from acute variceal hemorrhage (more than 72 h after the index bleed)

alcohol-associated and hepatitis B–related cirrhosis. Thus, whether these results can be generalized to patients with NASH-related cirrhosis remains to be determined.

Guidance statements

19. TIPS should not be performed for the primary prevention of variceal bleeding.
20. In patients with acute esophageal variceal hemorrhage, preemptive “early” TIPS with PTFE-covered stents within 72 h (ideally <24 h) of initial upper endoscopy should be considered in patients who meet any of the following criteria: Child-Pugh class C 10–13 points or Child-Pugh class B 8–9 points with active bleeding at initial endoscopy despite concomitant treatment with vasoactive agents.

Role of TIPS as salvage or rescue treatment for variceal bleeding

TIPS has been used for decades for the management of uncontrolled variceal hemorrhage (salvage TIPS) or in patients who fail standard therapy within the same admission (rescue TIPS) (Table 4). All published studies regarding salvage/rescue TIPS are observational, and most TIPS placed were using bare (uncovered) stents, which, as mentioned previously, are no longer considered standard of care. Six more recent studies^[102–107] suggested that salvage TIPS should be used for refractory bleeding, and four of these included patients who had a covered TIPS placed.^[102,104–106] The results are comparable among these reports, with control of bleeding in 80% to 100% of cases. However, 6-week mortality was high, ranging from 27% to 55%, with patients dying from liver failure, infection, and/or renal failure.

Thus, the indications for salvage/rescue TIPS should be balanced against futility. Factors associated with a significantly higher mortality are infection, renal failure, need for pressors, balloon tamponade, a high MELD score, a high APACHE II score, and a high Child-Pugh score.^[102–105] Few data are available regarding patients with Child-Pugh scores > 13.^[102,105] The 1-year mortality for this group may approach 100%, suggesting that salvage/rescue TIPS in these patients might be futile. A recent large multicenter retrospective study outlined that high lactate > 12 mmol/L and/or MELD score > 30 were significantly associated with a mortality greater than 90% at 6 weeks.^[106] In a nonrandomized study performed in patients with ACLF with uncontrolled variceal hemorrhage,

TIPS was associated with an improved survival.^[104] However, identification of candidates is key and needs to be further determined. The course of bleeding episode is likely another important factor, and it remains to be determined whether, for example, a rapid decrease/increase of lactate is predictive of outcomes. Generally, the indications of rescue TIPS should be balanced with the general therapeutic plans for these patients, specifically whether the patient is listed for transplant. Indeed, a small series showed that rescue TIPS could be a successful bridge to liver transplantation, even in those patients with severe liver disease (CTP score > 13).^[108]

Guidance statements

21. TIPS should be used as salvage treatment in patients in whom variceal bleeding cannot be controlled with medical and endoscopic therapy (failure of standard of care).
22. TIPS should be used as rescue treatment in patients with early recurrent variceal bleeding despite medical and endoscopic therapy (failure of standard of care).
23. In patients with a MELD score > 30, lactate > 12 mmol/L, or Child-Pugh > 13, salvage/rescue TIPS should not be used unless TIPS is a bridge to liver transplantation in the short term.

The role of TIPS in the long-term management after recovery from acute variceal bleeding

Per previous AASLD and European Association for the Study of the Liver Guidelines,^[8,89] the first-line therapy for the prevention of recurrent variceal hemorrhage is the combination of NSBBs (propranolol, nadolol, or carvedilol) and esophageal variceal ligation (EVL), with TIPS being the treatment of choice in patients who rebleed despite NSBB plus EVL. That is, TIPS is second-line therapy in the prevention of recurrent variceal bleeding. This is based on RCTs that, despite consistently showing effectiveness of TIPS in preventing rebleeding, have also shown an increased risk of hepatic encephalopathy without differences in survival.^[54,109]

Another group of patients in whom TIPS could be considered are those who recover from variceal bleeding and have PVT. In a recent trial including patients with thrombosis of > 50% of the portal vein trunk (complete in 30% of the patients), TIPS with anticoagulation was more effective than the combination of NSBB/EVL/anticoagulation in preventing variceal

rebleeding, without increasing the risk of hepatic encephalopathy and without an effect on survival.^[110] TIPS was also associated with a higher rate of portal vein recanalization, but again without improving survival as compared with standard therapy. Whether TIPS should be favored in patients with PVT is still uncertain. AASLD Guidelines on PVT recommend considering portal vein recanalization followed by TIPS in patients with chronic PVT and recurrent bleeding after failure of medical/endoscopic therapy.^[17] Therefore, similar to patients without PVT, TIPS should be used as second-line therapy in these patients.

Guidance statement

24. In patients who have recovered from an acute esophageal variceal hemorrhage and who did not have TIPS placed during admission, TIPS should be used as second-line treatment when rebleeding occurs despite adequate secondary prophylaxis with NSBBs and endoscopic therapy.

Role of TIPS and RTO in the management of gastrofundal varices (GOV2 and IGV1)

Prevalence of gastrofundal varices is lower than esophageal varices in patients with cirrhosis. In addition, the risk of bleeding from gastrofundal varices is lower than for esophageal varices.^[9] In patients considered for primary prophylaxis in a recent series, the bleeding rate was ~10% per year.^[111,112] Hemorrhage from gastrofundal varices tends to be more severe than from esophageal varices, and mortality in old series was as high as 45%.^[9] However, with currently available therapies, mortality is not different from mortality from esophageal variceal bleeding (~15% at 30 d).^[113] This should be taken into account when considering invasive treatments with narrow benefits-to-risk margins.

Primary prophylaxis of gastric varices: role of TIPS and RTO

The potential role of TIPS or RTO for primary prevention of gastric variceal hemorrhage has not yet been assessed in randomized trials. Results from trials of prophylactic surgical shunt therapy show a significantly higher rate of encephalopathy and a tendency for a higher mortality in patients randomized to shunt

surgery.^[114] Therefore, TIPS is not recommended for primary prophylaxis of gastric variceal bleeding. In patients treated with RTO for primary prophylaxis, nonrandomized studies suggest a high rate of variceal obliteration and low rate of bleeding but no effect on mortality as compared with no intervention.^[112]

Guidance statement

25. Neither TIPS nor RTO should be used to prevent first hemorrhage in patients with gastrofundal varices.

Acute bleeding from gastrofundal varices (GOV2 and IGV1)

General management of bleeding from gastrofundal varices is the same as that for esophageal variceal hemorrhage, as specified in the AASLD guidance on portal hypertension.^[1] In cases of massive bleeding, balloon tamponade (with Linton-Nachlas tube or Sengstaken-Blakemore tube with gastric balloon inflation) may serve as a bridge to other treatments. Endoscopy is performed for initial diagnosis of the source of bleeding. In centers with expertise, endoscopic therapy with cyanoacrylate can be attempted to stabilize the patient and might serve as definitive therapy.^[113,115–117] Patients should be promptly transferred to centers with expertise in the management of gastric variceal hemorrhage.

All patients with bleeding gastrofundal varices should undergo a comprehensive imaging workup because defining the specific anatomy of afferent and efferent vessels and the location of the nidus of the varices determines the optimal management of gastrofundal varices. Although infrequent, isolated splenic vein thrombosis/compression can cause segmental/left-sided portal hypertension, which will not benefit from TIPS. Splenectomy, splenic embolization, or trans-splenic ATO should be considered in these cases.

Role of TIPS in the management of acute gastrofundal variceal hemorrhage

Preemptive TIPS. As discussed in previous sections, patients with variceal hemorrhage and Child B with CTP of 8–9 points with active bleeding at initial endoscopy and Child C with CTP <14 points are at high risk of treatment failure and death and might benefit from preemptive TIPS. Preemptive TIPS trials included both GOV1 and

GOV2, although none of the reported patients had IGV. Still, the same rationale for a preemptive approach to prevent further rebleeding and ensuing deteriorating clinical status would apply to IGV, with the distinction that patients with GOV2 and IGV have alternative options to preemptive TIPS such as RTO.

Because TIPS or RTO (together with cyanoacrylate injection with or without NSBBs) can be used as first-line therapies for preventing rebleeding in these patients, the selection of patients at high risk of failure is less relevant. Two multicenter trials are currently ongoing in Spain (NCT02364297) and France (NCT03705078) in patients with GOV2/IGV1 initially treated with vasoactive drugs and cyanoacrylate injection, randomizing patients to preemptive TIPS within 3–5 days versus a control arm using the combination of repeated cyanoacrylate injection and NSBBs.

Salvage/rescue TIPS. TIPS is highly effective in achieving hemostasis in patients with uncontrolled gastroduodenal variceal bleeding (salvage TIPS) (87%–100% of control of bleeding).^[92,118] The same considerations for futility should be applied as in the case of esophageal variceal bleeding.

Prevention of rebleeding from gastroduodenal varices (GOV2/IGV1). Patients surviving an acute gastroduodenal variceal (GOV2/IGV1) bleeding episode after endoscopic treatment have a rebleeding rate that ranges from 7% to 65%; therefore, secondary prophylaxis should be initiated as soon as possible.^[119]

Only one RCT has compared TIPS (performed with bare stents) versus repeated cyanoacrylate injection in 72 patients for the prevention of gastroduodenal variceal rebleeding.^[120] In this trial, all patients had initial control of bleeding with cyanoacrylate. The study showed lower rates of rebleeding with TIPS, with higher encephalopathy rates and no differences in survival. Observational studies showed similar results.^[121,122] On this basis, TIPS is recommended among first-line therapies to prevent gastroduodenal variceal rebleeding by AASLD^[8] and other recent guidelines.^[8,23,24,123] However, trials that will define subgroups that will benefit from TIPS versus BRTO (compared with endoscopic and pharmacological therapy with NSBBs) are strongly needed.

ATO in combination with TIPS in patients with gastroduodenal variceal hemorrhage. Gastroduodenal varices represent a different scenario from esophageal varices because the competitive downhill gradient from splenic vein to renal vein is typically lower compared with that of esophageal varices, which drain through the coronary-azygous veins. In addition, GRS feeding gastric varices frequently have a larger diameter than the TIPS, which often results in persistent flow through varices despite achieving low post-TIPS portal pressure gradients. Thus, obliteration of residual gastric varices during TIPS may be important in patients treated for acute refractory gastric variceal bleeding. Several retrospective studies including a small number of cases with gastroduodenal varices suggest

an increased efficacy of combination therapy (TIPS + embolization) as compared with TIPS alone in preventing rebleeding.^[124–128] Additionally, a recent small RCT in patients with variceal hemorrhage, 82% from gastroduodenal varices, suggests that concurrent embolization of large spontaneous shunts at the time of TIPS placement decreases the risk of hepatic encephalopathy.^[37]

Small series have reported the use of ATO without TIPS for the management of gastric varices but showed high rebleeding rates (3.7%–36.7% and 21.6%–53.6% at 12 and 24 months).^[88,129] Although this strategy might be considered in patients with contraindications to TIPS, the invasiveness and the availability of alternatives (NSBBs, endoscopic therapy, and RTO) makes it unlikely that this will be evaluated in randomized trials and should only be considered as a nonstandard treatment in patients who are not candidates for first-line therapies.

The role of RTO in the management of gastroduodenal varices

In patients with a left-sided spontaneous portosystemic shunt (mainly gastroduodenal/gastrocaval shunt), obliteration of gastroduodenal varices by RTO may be a therapeutic alternative for the management of bleeding from gastroduodenal varices.^[68]

RTO in uncontrolled gastroduodenal variceal hemorrhage. There is almost no evidence regarding the role of salvage RTO in gastroduodenal variceal bleeding.^[130–132] The largest study available^[132] retrospectively evaluated 52 patients (50% GOV2 and 50% IGV) with treatment failure who underwent BRTO. Patients with concomitant EV also received endoscopic ligation. Authors showed a survival rate of 92% at 1 year, and only one patient required rescue treatment with TIPS. Because traditional BRTO is a long procedure, it is not ideal in patients with uncontrolled bleeding. CARTO or PARTO are preferable because they can be completed in a similar time frame as TIPS.

RTO in the prevention of gastric variceal rebleeding. Four nonrandomized studies, including a mix of patients on primary prophylaxis, acute bleeding, and secondary prophylaxis, directly compared cyanoacrylate injection with RTO, showing that RTO was more effective in the prevention of rebleeding from gastroduodenal varices.^[133–136] This was confirmed in a recent randomized trial comparing RTO with repeated cyanoacrylate injection in the prevention of gastric variceal bleeding in 64 patients.^[137] This trial showed a higher efficacy of RTO in preventing rebleeding (2 year probability of rebleeding 7% vs. 35% with cyanoacrylate) with no differences in survival, complications, or the rate of worsening esophageal varices.

Choice between TIPS or RTO. Both TIPS and RTO have been shown to be superior to cyanoacrylate injection in RCTs for the secondary prevention of gastric variceal hemorrhage in terms of efficacy without differences in survival. However, no RCT has compared

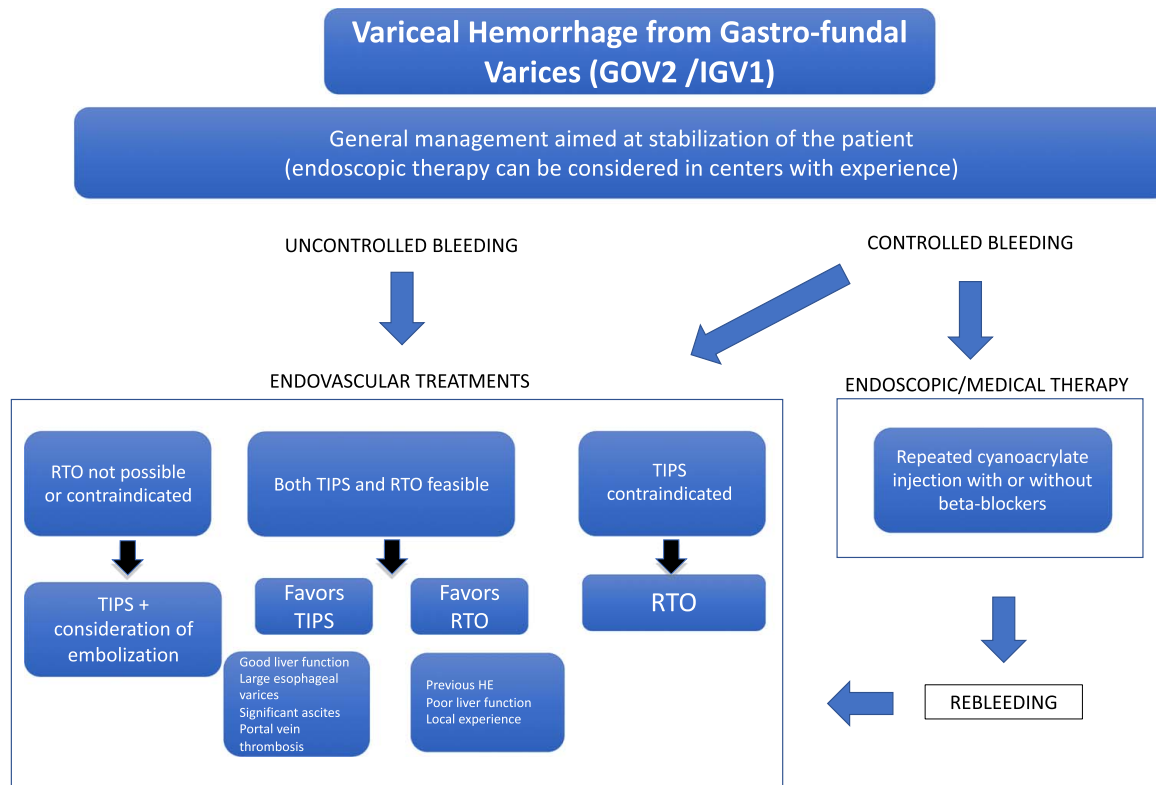


FIGURE 6 Role of TIPS and RTO in the management of bleeding from gastrofundal varices (GFV). After initial management, that might include endoscopic treatment in experienced centers, patients with uncontrolled bleeding should undergo TIPS or RTO. In patients in whom bleeding is initially controlled, either endovascular treatment (TIPS or RTO) or endoscopic/medical therapy (repeated cyanoacrylate injection with or without beta-blockers) might be used as first-line treatments to prevent rebleeding. Patients with rebleeding after endoscopic/medical therapy should undergo TIPS or RTO. Choice between TIPS or RTO depends on the anatomy, clinical profile of the patient, and local expertise.

TIPS versus RTO. In a recent meta-analysis of nonrandomized studies, both TIPS and RTO showed similar hemostasis rates and postoperative procedure-related complications.^[138] Rates of hepatic encephalopathy were higher in patients treated with TIPS, although uncovered stents were used in most of these studies.^[139–142]

Generally, the choice of RTO versus TIPS is determined by clinical and pathophysiological considerations and local expertise (Fig. 6). RTO closes a preexisting portosystemic shunt, whereas TIPS creates a new one; therefore, the main advantage of RTO over TIPS is the reduction in hepatic encephalopathy and potentially the prevention of liver function deterioration related to portal systemic shunting, whereas the downside of RTO is the increased risk of new/worsened esophageal varices and new/worsened ascites and/or hydrothorax. In addition, patients with cardiac contraindications for TIPS are commonly eligible for RTO.

Therefore, in patients with contraindications for TIPS, RTO should be the first choice, provided vascular anatomy and local expertise allow for its performance. In patients who are candidates for both TIPS and RTO, the choice might be based on local expertise and the presence of other complications of portal hypertension

(i.e., large esophageal varices, significant ascites or PVT that would favor TIPS, and history of hepatic encephalopathy or deteriorated liver function that would favor RTO).

Although it has been suggested that a combination of TIPS + RTO could be more efficacious in preventing both rebleeding and other portal hypertension complications,^[74] data are scarce to support this recommendation beyond individual cases.

Guidance statements

26. In patients with acute hemorrhage from gastrofundal varices (GOV2/IGV1), cyanoacrylate injection, TIPS, or RTO can be considered first-line therapies to control bleeding and prevent rebleeding.
27. RTO is the first-line endovascular option for hemorrhage from gastrofundal varices in patients with contraindications to TIPS.
28. In patients with hemorrhage from gastrofundal varices who are candidates for both

RTO and TIPS, the choice of one modality versus the other depends on vascular anatomy, clinical profile, and center expertise.

- RTO would be favored in patients with histories of overt hepatic encephalopathy or poor liver function.
 - TIPS would be favored in patients with other complications of portal hypertension, such as large esophageal varices or significant ascites, and in patients with PVT.
29. On a case-by-case basis, TIPS can be complemented with variceal embolization because this may reduce hepatic encephalopathy and rebleeding.

Endovascular treatments for ectopic varices

Ectopic varices are rare manifestations of cirrhotic portal hypertension. These abnormal vessels are found mostly in parts of the gastrointestinal tract such as duodeno-jejunal and colorectal areas. They can also appear in and around the intestinal stoma and very rarely in the urinary bladder wall, gallbladder, and bile duct, especially in patients with prior abdominal surgeries. The incidence of bleeding is low and depends on the location of these varices. However, when bleeding occurs, it may be massive and life-threatening.^[3] This guidance focuses on interventional radiology techniques for ectopic varices.

Unlike the well-established approaches to esophageal and gastric varices, there is little consensus on treatment of gastrointestinal ectopic varices. Endoscopic treatments include endoscopic variceal ligation (EVL), injection sclerotherapy, cyanoacrylate glue injection, thrombin injection, and glue-coil embolization. These have been reported with varying success, depending on the location of the varices and the technique employed. In general, they are associated with high risk of recurrence, rebleeding, and ulceration. More recently, many of these treatments have been performed under endoscopic US guidance, but the availability of this expertise is limited. Endoscopic therapy may be used as a bridge to more definitive therapy. In most instances, more definitive therapies are based on fluoroscopic guided transvenous approaches such as variceal embolization or sclerotherapy with or without pressure reduction with TIPS. Because of the rarity of these events, reported treatment outcomes are based on single-case reports or small retrospective case series. Accordingly, systematic analyses of such data yield limited evidence-based recommendations for choosing among the available therapeutic options for the different types of ectopic varices.^[143]

Nevertheless, there are general principles that can be used to guide our approach to ectopic varices. First, patients need to have a comprehensive imaging evaluation of the relevant vascular anatomy. Second, decompression with TIPS alone is often insufficient to control ectopic variceal bleeding because the ectopic vessel-systemic pressure gradient may be lower than the post-TIPS PSPG allowing for persistent downhill flow through the ectopic varices. This emphasizes the routine need for definitive concomitant embolization of ectopic varices by antegrade or retrograde techniques. Third, ATO alone may be sufficient in many cases when all relevant variceal feeders can be obliterated, in which case post-TIPS complications such as encephalopathy could be avoided. This scenario is distinct from gastroduodenal varices, which are frequently fed by large collaterals wherein control of bleeding is rarely achieved with antegrade embolization alone (without TIPS). Fourth, RTO or direct puncture and embolization-/obliteration may be feasible when favorable anatomy allows access to the varices. Both ATO and RTO are the preferred approaches in poor candidates for TIPS. These principles can allow adequate multidisciplinary planning among hepatologists, gastroenterologists, interventional radiologists, and surgeons.

Duodenal varices

Varices are rare in the duodenum; however, they can be the cause of severe bleeding, with mortality rates exceeding 40%.^[3] They are most commonly located in the duodenal bulb and second portion of the duodenum. TIPS with or without embolization and percutaneous ATO or RTO have shown high success rates in the control of duodenal variceal bleeding.^[143] The latter methods are preferable over TIPS in patients with advanced cirrhosis and hepatic dysfunction and in those with severe hepatic encephalopathy.^[144,145]

Jejunal/intestinal varices

The triad of presence of portal hypertension, previous abdominal surgery, and melena of unidentified source are usually the clinical features of small bowel variceal bleeding.^[146] In recent reports, ATO and RTO have been shown to effectively control variceal bleeding.^[147,148]

Rectal varices

Rectal varices can commonly be seen in patients with portal hypertension because of the connection between the tributaries of the inferior mesenteric vein and intrinsic rectal venous plexus. These are differentiated from hemorrhoids by their presence above the

dentate line. Bleeding is rare, but when it occurs, it can be massive, uncontrolled, and life-threatening.^[149,150] Like many other ectopic varices, there is no well-defined management protocol for treating bleeding rectal varices. Endoscopic therapies, such as band ligation, injection sclerotherapy, and coil embolization, especially when used in conjunction with endoscopic US and Doppler, have been associated with modest success^[151–153] and may be used as bridges to more definitive therapy or when interventional radiology therapies are not an option. Local surgical procedures like direct suturing and stapling of varices have very limited success in long-term control of bleeding and should be avoided.

TIPS with embolization is the primary and standard option for management of rectal varices. After an initial case report by Katz et al,^[154] two other larger series showed 67% and 78% long-term success rates in the control of bleeding rectal varices.^[155,156] In recent years, ATO and RTO alone have become viable alternative options for rectal varices, especially in patients with preexisting significant encephalopathy and/or poor hepatic reserve.^[157,158]

Stomal varices

Like other ectopic varices of the gastrointestinal tract, the chances of bleeding from these varices is low (3%–5%) but these bleeding episodes can be severe.^[159] Such varices can occasionally be difficult to identify by direct visualization or endoscopy, emphasizing the importance of cross-sectional contrast-enhanced CT or MR vascular imaging. Saad et al described two different presentations of stomal bleeding, one related to congestion leading to diffuse oozing and the other probably related to a discrete varix leading to focal bleeding.^[159] They concluded that the former responds better to TIPS and the latter is better controlled with ATO or RTO, with or without TIPS.

In a literature review of 163 patients with stomal variceal bleeding,^[7,160] 57 patients were treated with TIPS and 12 with surgical portosystemic shunt, 35 patients with endovascular embolization and RTO, 18 with surgical revision, and the rest with a variety of local therapies, including banding and sclerotherapy. With increased rates of recurrent bleeding with TIPS alone, a combination of these procedures has been found to associate with better results, emphasizing that TIPS should be associated with concomitant embolization or a combination of aforementioned procedures.

In summary, ectopic varices are rare and there is limited evidence-based cumulative experience regarding the best approaches to their management. It seems that, in most cases, after the initial endoscopic or imaging evaluation and potential local therapy, various transvenous approaches (ATO or RTO) associated with

potential portal decompression by TIPS are the preferred methods for the treatment of acute bleeding of ectopic varices and/or for the prevention of their recurrence. These decisions should be made on a case-by-case basis with a multidisciplinary team (hepatology, gastroenterology, interventional radiology, and surgery).

Guidance statements

30. Management of ectopic varices should be determined by a multidisciplinary team consisting of hepatologists, gastroenterologists, interventional radiologists, and surgeons.
31. Patients with suspected bleeding from ectopic varices should be assessed with comprehensive imaging to define the functional vascular anatomy of the varices, allowing for an individualized treatment approach.
32. Endoscopic therapy for ectopic varices, when feasible and available, may be used as a

TABLE 5 Areas of future research in endovascular treatments for variceal hemorrhage

1. Determining the best strategy for TIPS for variceal hemorrhage, including the optimal intraprocedural decrease in portal pressure gradient (based on source of bleed) and the best way to assess this gradient, including the internal zero (IVC vs. right atrium) and the timing of post-TIPS measurement
2. Assessing whether initial TIPS placement using a small (8 mm) diameter stent, independent of post-TIPS PSPG, will decrease post-TIPS complications while maintaining efficacy
3. Determining the outcomes of TIPS and portomesenteric thrombectomy versus anticoagulation in the treatment of variceal bleeding of patients with portomesenteric thrombosis
4. Testing the role of nonselective beta-adrenergic blockers as adjuvant therapy for TIPS, which could allow similar decreases in portal pressure with less portal systemic shunting and less complications
5. Determining the optimal technical approach to RTO
6. Addressing if TIPS results and complications in the treatment of variceal hemorrhage depend on etiology. Specifically, addressing if the results of TIPS trials conducted mainly in patients with alcohol-associated cirrhosis and viral-related cirrhosis can be extrapolated to patients with NASH-related cirrhosis
7. Updated large, randomized trials to clarify optimal patient selection for preemptive TIPS
8. Optimal therapy for primary prophylaxis of gastrofundal varices. The comparator should be NSBBs (or no therapy in patients with contraindications). Potential treatments for assessment could include cyanoacrylate injection, new endoscopic therapies (such as endoscopic US guided coil placement), TIPS, and RTO
9. Comparing efficacy, safety, and patient-related outcomes of TIPS versus RTO versus endoscopic therapy for the secondary prevention of gastric variceal hemorrhage in patients with a GRS

bridge to more definitive interventional radiology therapies.

33. TIPS is the first-line treatment for most ectopic varices and, in most cases, should be performed together with antegrade or retrograde transvenous obliteration (ATO/RTO) of the varices.
34. In patients who are poor candidates for TIPS or in patients in whom collaterals feeding varices are small, ATO or RTO alone (without TIPS) may provide sufficient control of ectopic variceal hemorrhage.

FUTURE RESEARCH AREAS

In the last few years, the refinement in the use of TIPS (mainly focused on patient selection and technical options) has led to an improvement in TIPS outcomes and expanded use. The challenges ahead include providing further evidence on the best technical approaches to TIPS, understanding the effects of TIPS in patients with NAFLD, now the fastest-increasing etiology of cirrhosis, and its therapeutic role as compared with emerging endovascular therapies, such as RTO. Table 5 summarizes priority areas for future research.

AUTHOR CONTRIBUTIONS

Conceptualization, Data curation, Methodology, Visualization, Writing – original draft, Writing – review & editing: All authors contributed equally. Project administration: Edward Wolfgang Lee, Juan G. Abraldes.

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CONFLICTS OF INTEREST

Ziv J. Haskal consults for Boston Scientific, Becton Dickinson, and WL Gore and Associates. He owns stock in Bendit, Blackswan Vascular, and Fluidx. Virginia Hernandez-Gea is on the speakers' bureau for Gore and Cook. Sanjeeva P. Kalva consults for Cannon, Boston Scientific, and Medtronic. Arpan Mohanty advises and received grants from Gilead. She received grants from Kinetix and NASHNET. Dominique Thabut is on the speakers' board for AbbVie, Alfasigma, Gilead, and Gore. Juan G. Abraldes received grants from Cook and Gilead. The remaining authors have no conflicts to report.

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